Stable Source Coding

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ISIT Presentation

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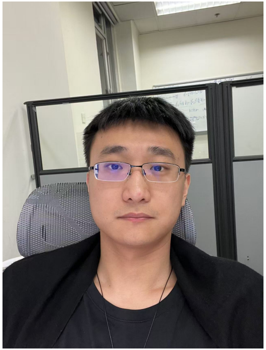

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Lossless source coding

$$X^n \longrightarrow \boxed{\text{Encoder } f_n} \longrightarrow f_n(X^n) \longrightarrow \boxed{\text{Decoder } g_n} \longrightarrow \hat{X}^n = g_n(f_n(X^n))$$

$$f_n : \mathcal{X}^n \to \{0,1\}^{\ell_n} \qquad \ell_n = \lfloor nR \rfloor \qquad g_n : \{0,1\}^{\ell_n} \to \mathcal{X}^n$$

$$P_e^{(n)} := \mathbb{P}[g_n(f_n(X^n)) \neq X^n] \, .$$

- If $R < H(X)$, then $P_e^{(n)}$ does not vanish.
- If $R > H(X)$, then $P_e^{(n)} \to 0$.

Random Binning

- Randomly assign each sequence $x^n \in \mathcal{X}^n$ to a binary bin label $f_n(x^n) \in \{0,1\}^{\ell_n}$.
- When $R > H(X)$, the typical set can be reliably recovered with high probability.

Motivation: Compression mappings are often discontinuous

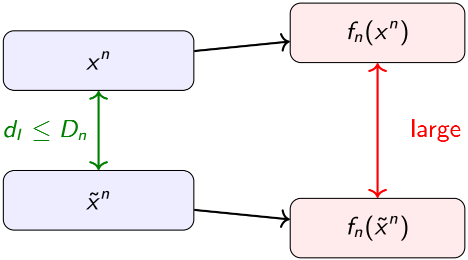
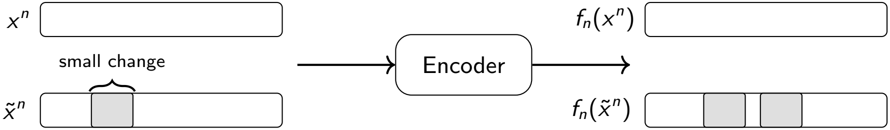


Figure: No local continuity structure.

Classical source coding

- Random binning assigns nearby sequences to unrelated bin indices.
- Vector quantization has cell boundaries: a tiny perturbation near the boundary can change the index drastically.

Motivation: Local update efficiency



$$d_I(x^n, \tilde{x}^n) \text{ small} \quad \Longrightarrow \quad d_{\mathrm{H}}(f_n(x^n), f_n(\tilde{x}^n)) \text{ small}$$

Local update efficiency [1, 2, 3]

If a stored source sequence needs to be updated locally,

$$d_I(x^n, \tilde{x}^n) \leq D_n,$$

then a stable encoder guarantees that only a bounded number of compressed bits must change.

Motivation: Neural compression

Neural compression viewpoint

Neural compressors often implement mappings with limited sensitivity to input perturbations, whereas optimal information-theoretic encoders may be highly discontinuous [4, 5, 6, 7].

We study the information-theoretic limits of imposing such local stability.

Question

What rates are achievable if the encoder must be locally stable? Is $R = H(X)$ still achievable?

Overview

Roadmap

Problem setup: Stable lossless source coding

Let $X_1, X_2, \ldots$ be an i.i.d. source on a finite alphabet $\mathcal{X}$. For blocklength n and rate R, the encoder and decoder are defined as:

$$f_n : \mathcal{X}^n \to \{0,1\}^{\ell_n}, \qquad g_n : \{0,1\}^{\ell_n} \to \mathcal{X}^n, \qquad \ell_n = \lfloor nR \rfloor.$$

(D_n, D'_n)-stability

The encoder f_n is (D_n, D'_n)-*stable* if, for all $x^n, \tilde{x}^n \in \mathcal{X}^n$, under input metric d_I and Hamming distance d_{H}:

$$d_I(x^n, \tilde{x}^n) \le D_n \quad \Longrightarrow \quad d_{\mathrm{H}}\big(f_n(x^n), f_n(\tilde{x}^n)\big) \le D'_n.$$

Achievability

A rate R is (D_n, D'_n)-*achievable* if there exist codes with

$$P_e^{(n)} := \mathbb{P}\big[g_n(f_n(X^n)) \neq X^n\big] \to 0, \qquad f_n \text{ is } (D_n, D'_n)\text{-stable}.$$

Two regimes on D_n and D'_n

Linear regime

$$D_n = \Theta(n), \quad D'_n = \Theta(n),$$

$$d_1 = \lim_{n\to\infty} \frac{D_n}{n}, \qquad d_2 = \lim_{n\to\infty} \frac{D'_n}{n}.$$

Typical nontrivial range: $d_1/2 \leq p$ and $d_2 \leq R$.

Sublinear regime

$$D_n = o(n), \qquad D'_n = o(n).$$

important special case: D_n, D'_n constants.

For the main bounds, take $\mathcal{X} = \{0, 1\}$, $P_X(1) = p \in (0, 1/2)$, and $d_I = d_{\mathrm{H}}$.

Roadmap

1. Problem Setup
2. Literature review: existing works and discussions
3. Intuition and proof via graph homomorphisms
4. Our results
5. Simulations and plots

Closest existing model: Smooth compression

Montanari–Mossel smooth compression [8]

Montanari and Mossel impose an L-Lipschitz smoothness constraint

$$d_{\mathrm{H}}\big(f_n(x^n), f_n(\tilde{x}^n)\big) \leq L\, d_I(x^n, \tilde{x}^n), \qquad \forall x^n, \tilde{x}^n \in \mathcal{X}^n.$$

Theorem II.1 in [8]: *For a dense class of source distributions, there exists a bounded (independent of n) but sufficiently large L such that $R = H(X)$ is still achievable under the L-smoothness constraint.*

Our viewpoint

- We allow two independent scales and only consider *local* pairs:

$$\forall x^n, \tilde{x}^n \text{ s.t. } d_I(x^n, \tilde{x}^n) \leq D_n \quad \Longrightarrow \quad d_{\mathrm{H}}\big(f_n(x^n), f_n(\tilde{x}^n)\big) \leq D'_n.$$

- Their condition implies (D_n, LD_n)-stability for every sequence $\{D_n\}$. Our setup is more general.
- Our degree bound is closely related to their converse in the binary case.
- Our clique-size bound gives a generally stronger converse in the simulations.

Related ideas and what is different here

Local update / decode efficiency [1, 2, 3]

Vatedka et al. consider a bounded (maximum) number of codeword bits that must be read or written after one source bit update.

Theorem II.1 in [1]: *For Bernoulli*(p) *sources,* $\epsilon > 0$*, there exists a compression scheme with* $R = H(p) + \epsilon$*, such that*

$$d_{\mathrm{H}}(x^n, \tilde{x}^n) = 1 \quad \Longrightarrow \quad d_{\mathrm{H}}(f_n(x^n), f_n(\tilde{x}^n)) \leq O\left(\tfrac{1}{\epsilon} \log \log n\right).$$

In our language, this says $R = H(p) + \epsilon$ is $(D_n, O(\frac{1}{\epsilon} D_n \log \log n))$-achievable.

Major differences from our problem setup

- Their setup also constrains the number of bits that must be read to perform a local update.
- $(D_n, D_n' = O(\frac{1}{\epsilon} D_n \log \log n))$ does not fall into our main linear-regime.
- Their result is also not sharp enough for our sublinear-regime setting. For example, when $D_n = O(1)$, their bound only gives $D_n' = O(\log \log n)$, rather than a constant D_n'.

Roadmap

Overview of our proof

Step 1: Reduction

Original problem

$X^n \overset{\text{i.i.d.}}{\sim} P_X$

$P_e^{(n)} \to 0$

→

Converted problem

$X^n \sim \mathrm{Unif}(\mathcal{A}_n),\ \mathcal{A}_n \subseteq \mathcal{T}(\pi_n)$

$P_e^{(n)} = 0$ on $\mathcal{A}_n$, $\dfrac{|\mathcal{A}_n|}{|\mathcal{T}(\pi_n)|} \to 1$

Step 2: Graph formulation

Stability condition

$d_{\mathrm{H}}(x^n, \tilde{x}^n) \le D_n$

$\Rightarrow d_{\mathrm{H}}(f_n(x^n), f_n(\tilde{x}^n)) \le D_n'$

→

Injective graph homomorphism

$G_n' \xrightarrow{f_n} H_n$

Source graph: $G_n' := G_n[\mathcal{A}_n]$,

Codeword graph: H_n.

Step 3: Converse bounds

Necessary conditions from $G_n' \xrightarrow{f_n} H_n$

Maximum degree: $\Delta(G_n') \le \Delta(H_n)$

Maximum clique: $\omega(G_n') \le \omega(H_n)$

→ Degree bound

→ Clique bound

Step 1: Strong typicality

Let $\hat{P}_{x^n}$ denote the empirical distribution of x^n, and choose

$$\epsilon_n \to 0, \qquad \epsilon_n \sqrt{n} \to \infty.$$

Define the set of strongly typical n-types by

$$\mathcal{P}_n(\epsilon_n) \triangleq \Big\{ \pi_n \in \mathcal{P}_n(\mathcal{X}) : \forall a \in \mathcal{X}, |\pi_n(a) - P_X(a)| < \epsilon_n \Big\},$$

where $\mathcal{P}_n(\mathcal{X})$ is the set of all types with denominator n.
For each type $\pi_n \in \mathcal{P}_n(\mathcal{X})$, define the corresponding type class

$$\mathcal{T}(\pi_n) \triangleq \{ x^n \in \mathcal{X}^n : \hat{P}_{x^n} = \pi_n \}.$$

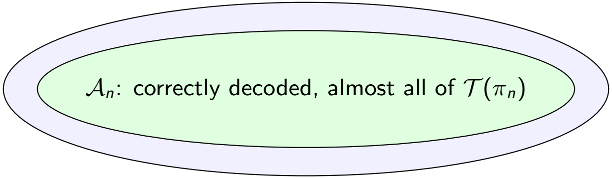
Step 1: Exact decoding on a large single-type subset

Choice of one type class

For any sequence of reliable codes, there exists a sequence of strongly typical types $\pi_n \in \mathcal{P}_n(\epsilon_n)$ and subsets $\mathcal{A}_n \subseteq \mathcal{T}(\pi_n)$ such that

$$\frac{|\mathcal{A}_n|}{|\mathcal{T}(\pi_n)|} \to 1, \qquad g_n(f_n(x^n)) = x^n, \quad \forall x^n \in \mathcal{A}_n.$$

$$\mathcal{T}(\pi_n) = \{x^n : \hat{P}_{x^n} = \pi_n\}$$



Replace a high-probability statement by an exact statement on a large "good" set, then prove a deterministic combinatorial statement on that "good" set.

Step 2: Define source and codeword neighborhood graphs

Source graph G_n

$$V(G_n) = \mathcal{T}(\pi_n),$$

with an edge if

$$d_{\mathrm{H}}(x^n, \tilde{x}^n) \leq D_n.$$

For binary sources, this is a union of *generalized Johnson graphs* on a constant-weight slice.

Codeword graph H_n

$$V(H_n) = \{0,1\}^{\ell_n},$$

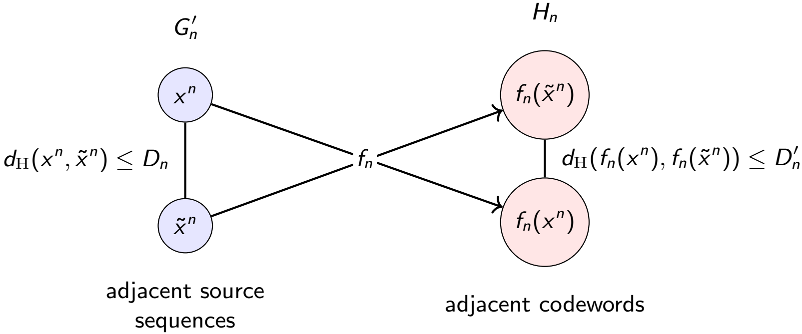
with an edge if

$$d_{\mathrm{H}}(f_n(x^n), f_n(\tilde{x}^n)) \leq D'_n.$$

This is the D'_n-th power of the ℓ_n-dimensional *Hamming graph*.

Step 2: Stability becomes graph homomorphism

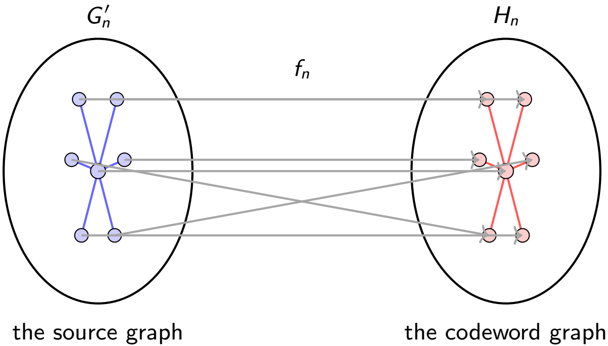
Restrict G_n to the correctly decoded set $\mathcal{A}_n$ and denote the induced graph by G'_n.



Injective graph homomorphism

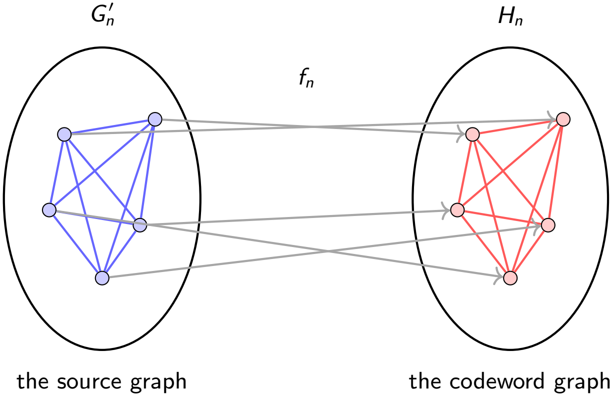
Exact decoding on $\mathcal{A}_n$ implies that f_n is injective on $\mathcal{A}_n$. $G'_n \xrightarrow{f_n} H_n$ is an injective graph homomorphism.

Step 3: Graph homomorphism–Degree



A necessary condition: *the maximum degree of H_n is no less than the maximum degree of G'_n.*

Step 3: Graph homomorphism–Clique



A necessary condition: *the maximum clique size of H_n is no less than the maximum clique size of G'_n.*

- The maximum degrees and maximum clique sizes are tractable for every n. For simplicity, in our main results we let $n \to \infty$.

Roadmap

Linear-regime degree bound

Define

$$s(x) = ph_2\left(\frac{x}{p}\right) + (1-p)h_2\left(\frac{x}{1-p}\right).$$

$$F(d_1, p) = \begin{cases} s(d_1/2), & 0 \le d_1/2 \le p(1-p), \\ h_2(p), & p(1-p) \le d_1/2, \end{cases}$$

$$G(d_2, R) = \begin{cases} Rh_2(d_2/R), & 0 \le d_2 < R/2, \\ R, & d_2 \ge R/2. \end{cases}$$

Theorem: degree bound

For every (D_n, D'_n)-achievable rate R with $D_n/n \to d_1 \le 2p$ and $D'_n/n \to d_2$,

$$\boxed{F(d_1, p) \le G(d_2, R).}$$

Linear-regime clique-size bound

Let

$$\Psi(d_2, R) = \begin{cases} Rh_2\left(\dfrac{d_2}{2R}\right), & 0 \le d_2 < R, \\ R, & d_2 \ge R. \end{cases}$$

Theorem: clique-size bound

For every (D_n, D_n')-achievable rate R with $D_n/n \to d_1 \le 2p$ and $D_n'/n \to d_2$,

$$\boxed{h_2(d_1/2) \le \Psi(d_2, R).}$$

This bound is independent of p except through the feasibility condition $d_1/2 \le p$.

Sublinear-regime converse (from degree)

For $D_n = o(n)$ and $D'_n = o(n)$, every stably decodable binary code must satisfy

$$\limsup_{n\to\infty} n^{D_n - D'_n} \frac{D_n'^{D'_n} \big((p-\epsilon_n)(1-p-\epsilon_n)\big)^{D_n/2}}{(D_n/2)^{D_n}(e^2R)^{D'_n}} \leq 1,$$

where $\epsilon_n \to 0$ and $\epsilon_n\sqrt{n} \to \infty$.

Constant-radius corollary

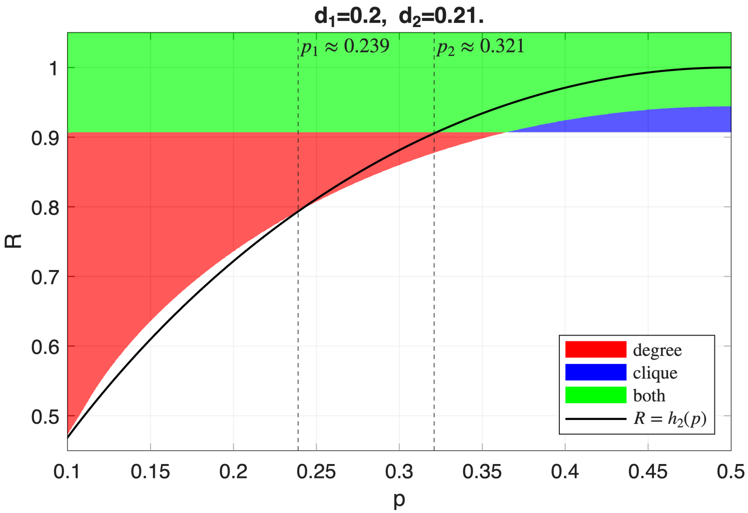
If $(D_n, D'_n) = (D, D')$ for constants D, D', then

$$\boxed{D \leq D'.}$$

This remains nontrivial even when $R > 1$, i.e., when the representation expands.

Roadmap

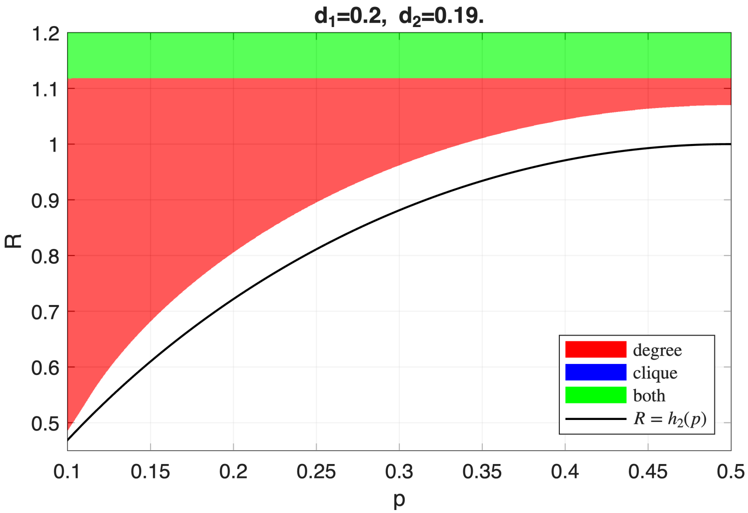
Numerical evaluation in the (p, R) plane



$d_1 = 0.2,\ d_2 = 0.21$

- Green: both converse inequalities are satisfied.
- Red: degree bound only.
- Blue: clique bound only.
- Degree becomes dominated by $R \geq h_2(p)$ after $p_1 \approx 0.239$.
- Clique remains active until $p_2 \approx 0.321$.
- Empirically, the degree bound is dominated by the clique bound and the entropy bound.

When $d_2 < d_1$, the clique bound dominates



$d_1 = 0.2$, $d_2 = 0.19$

- The clique-size bound is the dominant new restriction in the displayed range.
- Empirically, the degree bound is dominated by the clique bound.

Conclusions

Main message

Stability-constrained source coding can be analyzed via graph homomorphism.

1. Stability induces an injective homomorphism from a source neighborhood graph to a codeword neighborhood graph.
2. Degree and clique-size comparisons yield explicit converse bounds.
3. The Shannon entropy $H(X)$ is not achievable under some stability conditions.
4. In the binary linear regime, the clique-size bound and the entropy bound are often stronger than the degree bound.

Acknowledgments

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Thank you!

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